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Assignment -5 Artificial Intelligence

Q1. Minimax Search Algorithm (Case Study)

A gaming company is designing an AI for a two-player strategy game similar to chess. The AI must always choose the optimal move assuming the opponent is also playing optimally.

Tasks:

Explain the Minimax algorithm and how it works in adversarial search.

Represent the game as a game tree and show how Minimax evaluates nodes.

Discuss the role of utility functions and depth limits.

Analyze time and space complexity.

Suggest improvements for handling large game trees.

Ans.

- Concept: Used in two-player games. MAX maximizes score, MIN minimizes it.

- Working:

Build game tree then evaluate leaves then propagate values upward.

MAX picks highest, MIN picks lowest.

- Utility Function:

Assigns values (win = +1, loss = -1, draw = 0).

- Depth Limit:

Used to cut large trees; apply heuristic evaluation.

- Complexity:

Time = $O(bd)$, Space = $O(bd)$

- Improvement:

Use Alpha-Beta pruning, heuristics.

Q2. Alpha-Beta Pruning (Case Study)

The same gaming company wants to improve the performance of its AI because Minimax is too

slow for deep searches.

Tasks:

Explain how Alpha-Beta Pruning optimizes Minimax.

Illustrate pruning with a suitable example tree.

Compare performance with and without pruning.

Discuss best-case and worst-case scenarios.

Explain how move ordering affects efficiency.

Ans.

- Idea: Skip branches that won't affect final result.

Prune when $\alpha \geq \beta$

- Effect: Same result as Minimax, fewer nodes explored.

Performance:

- Best: $O(bd/2)$

- Worst: $O(bd)$

- Move Ordering:

Better ordering then more pruning then faster search.

Q3. Expert System (Case Study)

A hospital wants to build an AI-based system that can assist doctors in diagnosing diseases based on symptoms and patient history.

Tasks:

Define an Expert System and its components (knowledge base, inference engine, user interface).

Explain rule-based reasoning (forward and backward chaining).

Design a simple rule-based system for disease diagnosis.

Discuss advantages and limitations of expert systems in healthcare.

Suggest improvements using modern AI techniques.

Ans.)

- Definition: AI system that mimics

human expert decisions.

- Components: Knowledge base + Inference engine + UI
- Reasoning: Forward chaining (data then result)

Backward chaining (goal then data)

- Example Rule:

IF fever AND cough $\rightarrow$ Flu

- Pros: Fast, consistent
- Cons: No learning, hard to update
- Improvement: Add ML, fuzzy logic, probabilistic models

Q4. Swarm Intelligence System (Case Study)

A smart city project aims to optimize traffic flow using decentralized decision-making inspired by nature.

Tasks:

Define Swarm Intelligence and its characteristics.

Explain how decentralized agents can solve complex problems.

Compare swarm intelligence with traditional centralized systems. Provide examples such as particle swarm optimization or bird flocking. Discuss real-world applications in traffic and robotics.

Ans.

- Definition: Collective behavior of decentralized agents.

- Features:

Self-organized, local interaction, no central control

- Comparison:

Swarm $\rightarrow$ scalable, robust

Centralized $\rightarrow$ less flexible

- Examples: PSO, bird flocking

- Applications: Traffic control, robotics, routing

Q5. Ant Colony System (Case Study)

A delivery company wants to find the shortest route for multiple delivery

vehicles across a city
with dynamic traffic conditions.

Tasks:

Explain the Ant Colony Optimization (ACO) algorithm.

Describe how ants use pheromone trails to find optimal paths.

Model the delivery problem using ACO.

Compare ACO with traditional algorithms like Dijkstra.

Discuss advantages, limitations, and real-world applications

Ans.

- Concept: Ants find shortest path using pheromones.

- Working:

Shorter paths then stronger pheromone then more ants follow

- Model:

Nodes = locations, ants = vehicles, goal = shortest route

- ACO vs Dijkstra:

ACO: adaptive, dynamic